Тема: Настройка протокола STP (IEEE 802.1D)

Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

Полезная информация: схема сохранена на сервере в проекте под именем Menzhulin-lab2-template, можно использовать кнопку Duplicate

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров

настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл

4) При помощи wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет

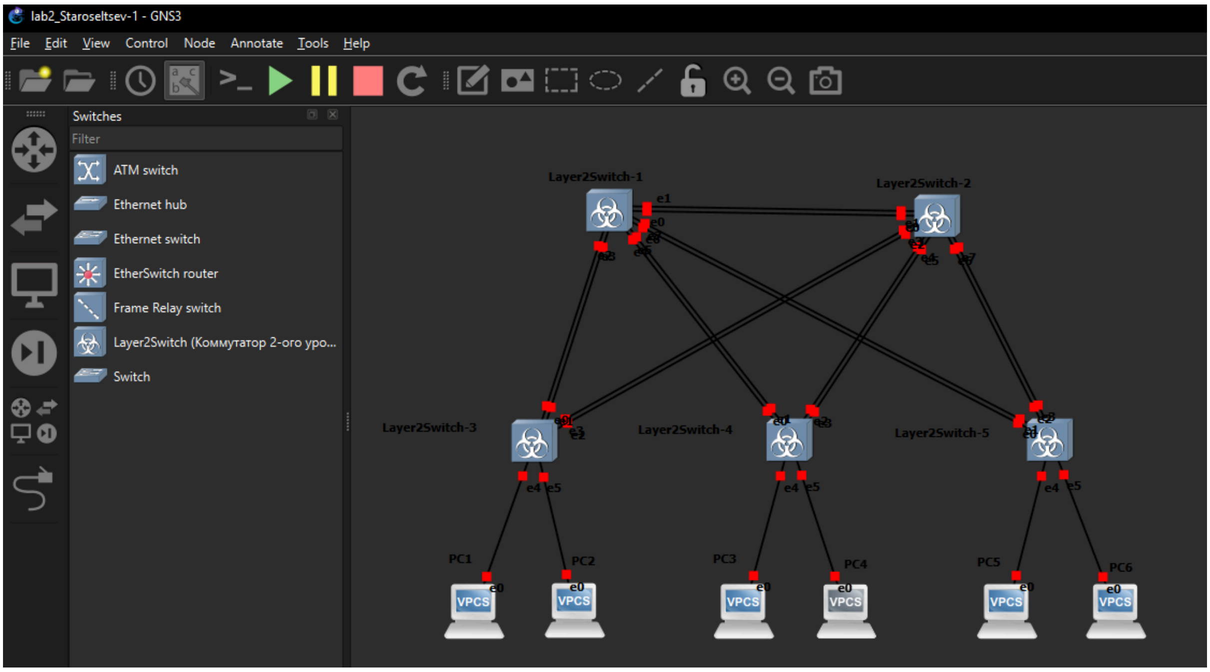
5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл

6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

7*) Опциональное задание: заменить STP на RSTP (IEEE 802.1w), повторить 1-6, отметить резервные порты в п.3 и п.5,

отличие работы протокола RSTP от протокола STP в п.4

1. Сделал дубликат схемы



Запустил все узлы

И на каждом коммутаторе выполнил команды

enable

show spanning-tree vlan 1

SW1

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c1d.3758.0000
             Cost        4
             Port        7 (GigabitEthernet1/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c21.59bb.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Root FWD 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

SW2

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c1d.3758.0000
             Cost        4
             Port        7 (GigabitEthernet1/2)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cbe.d2d7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------------------------
Gi0/0               Altn BLK 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Root FWD 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

SW3

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan1
                                  ^
% Invalid input detected at '^' marker.

vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c1d.3758.0000
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.f8b7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

SW4

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c1d.3758.0000
             Cost        8
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c2e.6f97.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

SW5

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0c1d.3758.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c1d.3758.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

По выводам команды, можно определить, что SW5 является root'ом
Я решил оптимальнее использовать SW1 или SW2 для построекния дерева, поэтому на SW1
enable
configure terminal
spanning-tree vlan 1 priority 0
end
write memory

```
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#spanning-tree vlan 1 priority 0
vIOS-L2-01(config)#end
vIOS-L2-01#write me
*May 26 10:03:06.940: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5274 bytes to 2022 bytes[OK]
*May 26 10:03:15.855: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 10:03:16.704: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
```

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0c21.59bb.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15  sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Desg FWD 4         128.7    Shr
Gi1/3               Desg FWD 4         128.8    Shr
```

SW5 перестал быть root'ом

```
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c1d.3758.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

2. Установил для всех компьютеров айпи адреса и сохранил конфигурацию

ip 192.168.1.i/24 ,i = 1,2,3,4,5,6

save

PC1

```
PC1> ip 192.168.1.1/24
Checking for duplicate address...
PC1 : 192.168.1.1 255.255.255.0

PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.1/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 21812
RHOST:PORT  : 127.0.0.1:21813
MTU         : 1500

PC1> save
Saving startup configuration to startup.vpc
.  done
```

PC2

```
PC2> ip 192.168.1.2/24
Checking for duplicate address...
PC2 : 192.168.1.2 255.255.255.0

PC2> save
Saving startup configuration to startup.vpc
.  done
```

PC3

```
PC3> ip 192.168.1.3/24
Checking for duplicate address...
PC3 : 192.168.1.3 255.255.255.0

PC3> save
Saving startup configuration to startup.vpc
.  done
```

PC4

```
PC4> ip 192.168.1.4/24
Checking for duplicate address...
PC4 : 192.168.1.4 255.255.255.0

PC4> save
Saving startup configuration to startup.vpc
.  done
```

PC5

```
PC5> ip 192.168.1.5/24
Checking for duplicate address...
PC5 : 192.168.1.5 255.255.255.0

PC5> save
Saving startup configuration to startup.vpc
.  done
```

PC6

```
PC6> ip 192.168.1.6/24
Checking for duplicate address...
saPC6 : 192.168.1.6 255.255.255.0

PC6> save
Saving startup configuration to startup.vpc
.  done
```

Проверка соединения компьютеров:
PC1
ping 192.168.1.2
ping 192.168.1.3
ping 192.168.1.4
ping 192.168.1.5
ping 192.168.1.6

```
PC1> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=12.354 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=4.156 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=0.892 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=3.052 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=1.756 ms

PC1> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=21.121 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=7.214 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=7.055 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=14.522 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=12.164 ms

PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=10.935 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=11.639 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=8.349 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=8.305 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=30.543 ms

PC1> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=11.382 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=15.293 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=10.786 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=10.320 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=13.974 ms

PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=7.319 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=6.191 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=18.775 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=7.650 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=18.089 ms
```

PC2
ping 192.168.1.1
ping 192.168.1.3
ping 192.168.1.4
ping 192.168.1.5
ping 192.168.1.6

```
PC2> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=5.561 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=6.040 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=3.211 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=8.212 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=7.512 ms

PC2> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=21.295 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=14.328 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=4.996 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=11.311 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=10.420 ms

PC2> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=12.724 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=14.788 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=12.648 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=26.413 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=14.699 ms

PC2> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=19.381 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=14.302 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=22.335 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=15.903 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=19.340 ms

PC2> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=15.897 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=25.355 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=17.705 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=5.150 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=10.130 ms
```

PC3
ping 192.168.1.1
ping 192.168.1.2
ping 192.168.1.4
ping 192.168.1.5
ping 192.168.1.6

```
PC3> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=24.039 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=13.310 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=20.943 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=17.278 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=4.468 ms

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=21.686 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=5.979 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=20.630 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=9.619 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=10.527 ms

PC3> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=6.571 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=7.199 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=1.357 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=2.091 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=2.653 ms

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=29.586 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=15.538 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=3.310 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=24.377 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=11.400 ms

PC3> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=12.518 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=20.026 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=15.980 ms
^C
PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=13.659 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=9.949 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=14.392 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=7.329 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=21.776 ms
```

PC4
ping 192.168.1.1
ping 192.168.1.2
ping 192.168.1.3
ping 192.168.1.5
ping 192.168.1.6

```
PC4> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=13.438 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=14.505 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=26.964 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=8.941 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=14.006 ms

PC4> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=22.110 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=20.885 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=15.078 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=15.070 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=19.098 ms

PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=10.362 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=5.919 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=2.362 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=3.958 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=9.537 ms

PC4> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=14.628 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=6.277 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=17.242 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=17.810 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=15.000 ms

PC4> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=17.877 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=14.180 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=7.413 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=5.311 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=12.963 ms
```

PC5

ping 192.168.1.1

ping 192.168.1.2

ping 192.168.1.3

ping 192.168.1.4

ping 192.168.1.6

```
PC5> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=8.216 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=24.605 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=10.143 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=11.706 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=12.193 ms

PC5> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=13.181 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=5.340 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=8.621 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=9.563 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=11.534 ms

PC5> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=12.880 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=11.079 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=16.209 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=10.093 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=15.478 ms

PC5> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=24.108 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=18.131 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=23.849 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=11.886 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=7.841 ms

PC5> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=2.352 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=2.489 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=3.390 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=2.917 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=9.233 ms
```

PC6
ping 192.168.1.1
ping 192.168.1.2
ping 192.168.1.3
ping 192.168.1.4
ping 192.168.1.5

```
PC6> ping 192.168.1.1

84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=16.849 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=16.021 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=7.636 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=3.602 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=13.083 ms

PC6> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=7.214 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=21.350 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=15.064 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=10.423 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=16.090 ms

PC6> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=12.113 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=17.867 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=17.948 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=18.056 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=8.114 ms

PC6> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=25.488 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=16.189 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=6.516 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=9.876 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=14.902 ms

PC6> ping 192.168.1.5

84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=3.180 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=9.487 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=7.718 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=4.770 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=7.926 ms
```

Проверка предыдущих соединений конечно избыточна, но я решил проверить все, чтобы не запутаться.

Соединение между компьютерами есть.

3. Повторно запустил show spanning-tree vlan 1 на всех свичах

SW1 (root)

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0c21.59bb.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- -------------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Desg FWD 4         128.7    Shr
Gi1/3               Desg FWD 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

SW2

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cbe.d2d7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Altn BLK 4         128.5    Shr
Gi1/1               Altn BLK 4         128.6    Shr
Gi1/2               Altn BLK 4         128.7    Shr
```

SW3

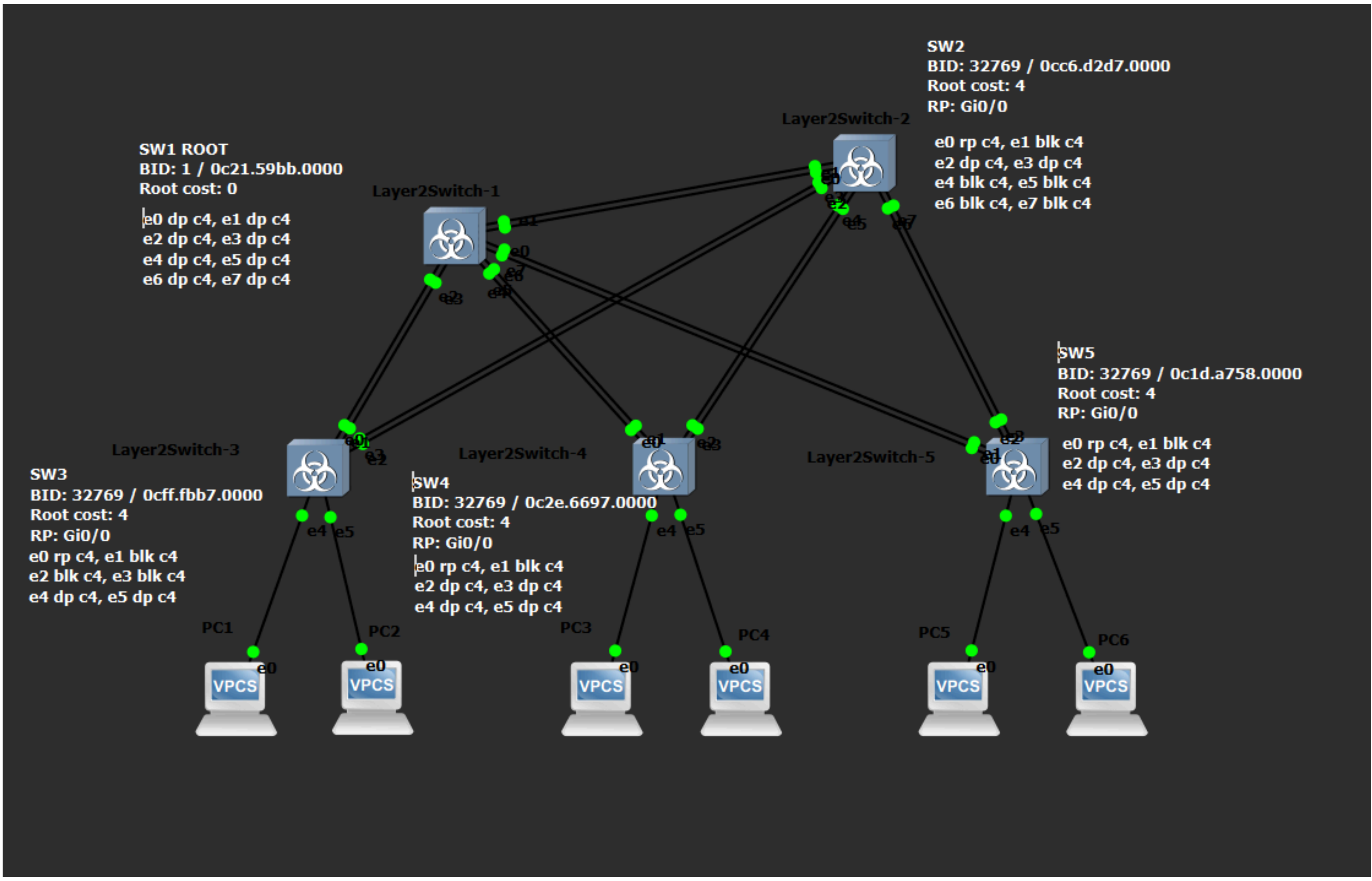
```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.f8b7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

SW4

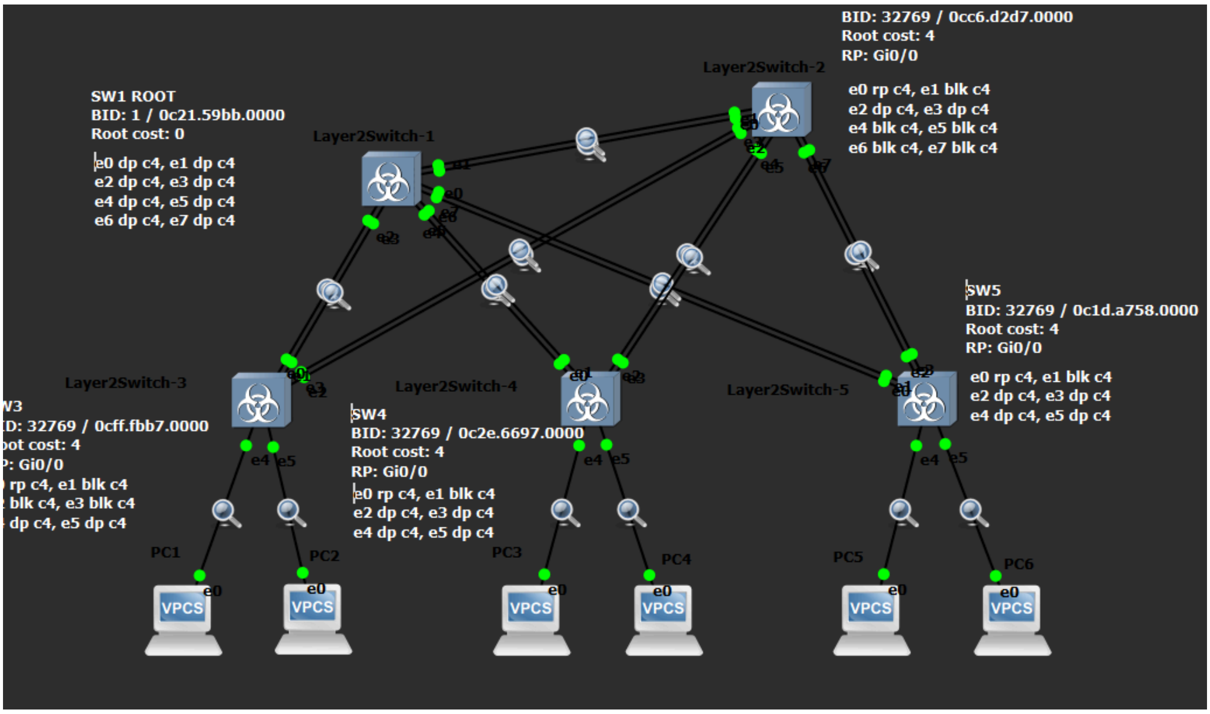
```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c2e.6f97.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

SW5

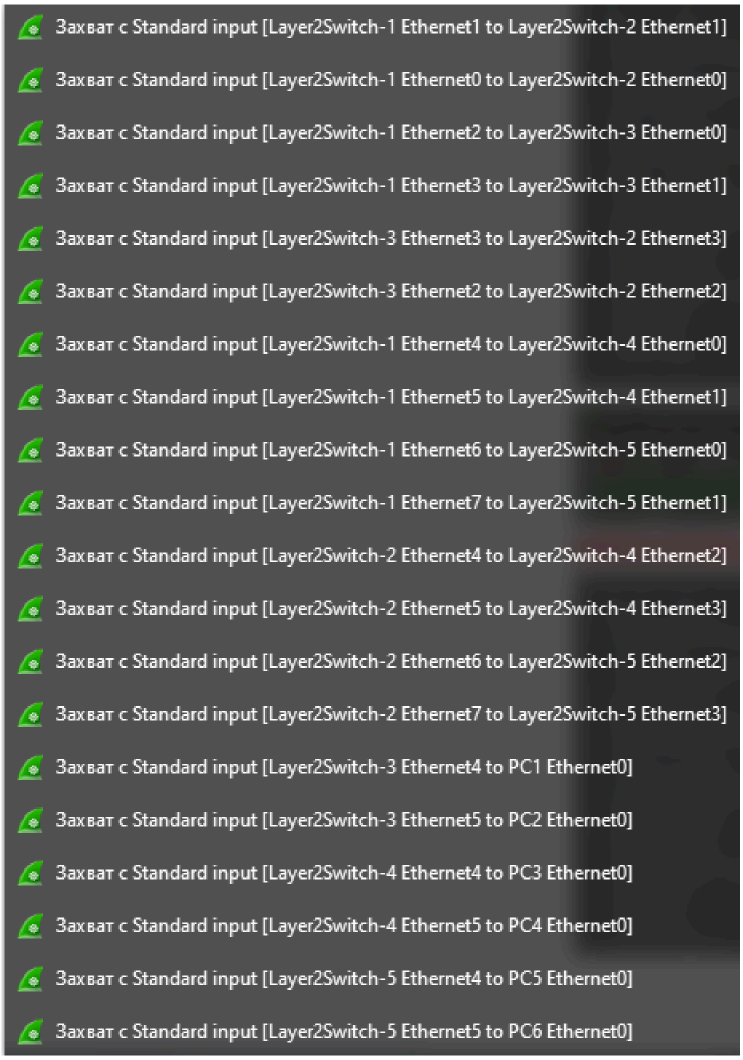
```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c1d.3758.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

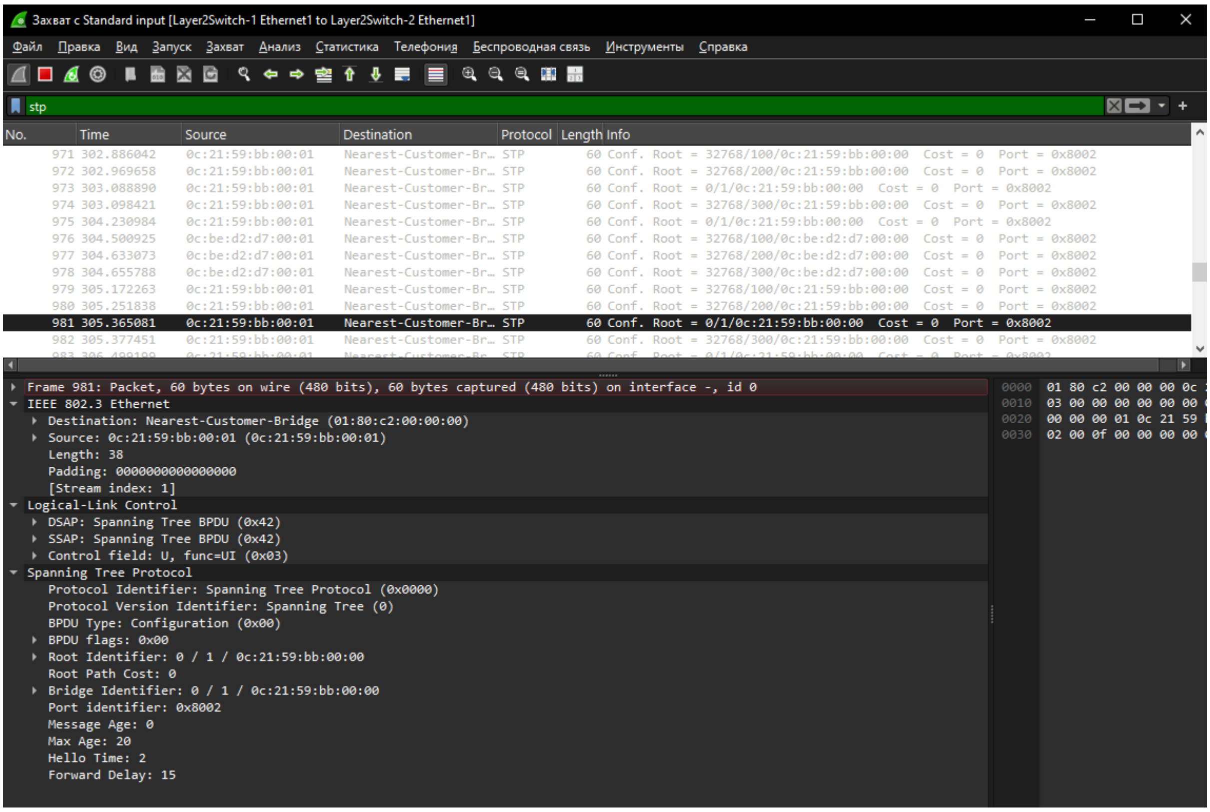


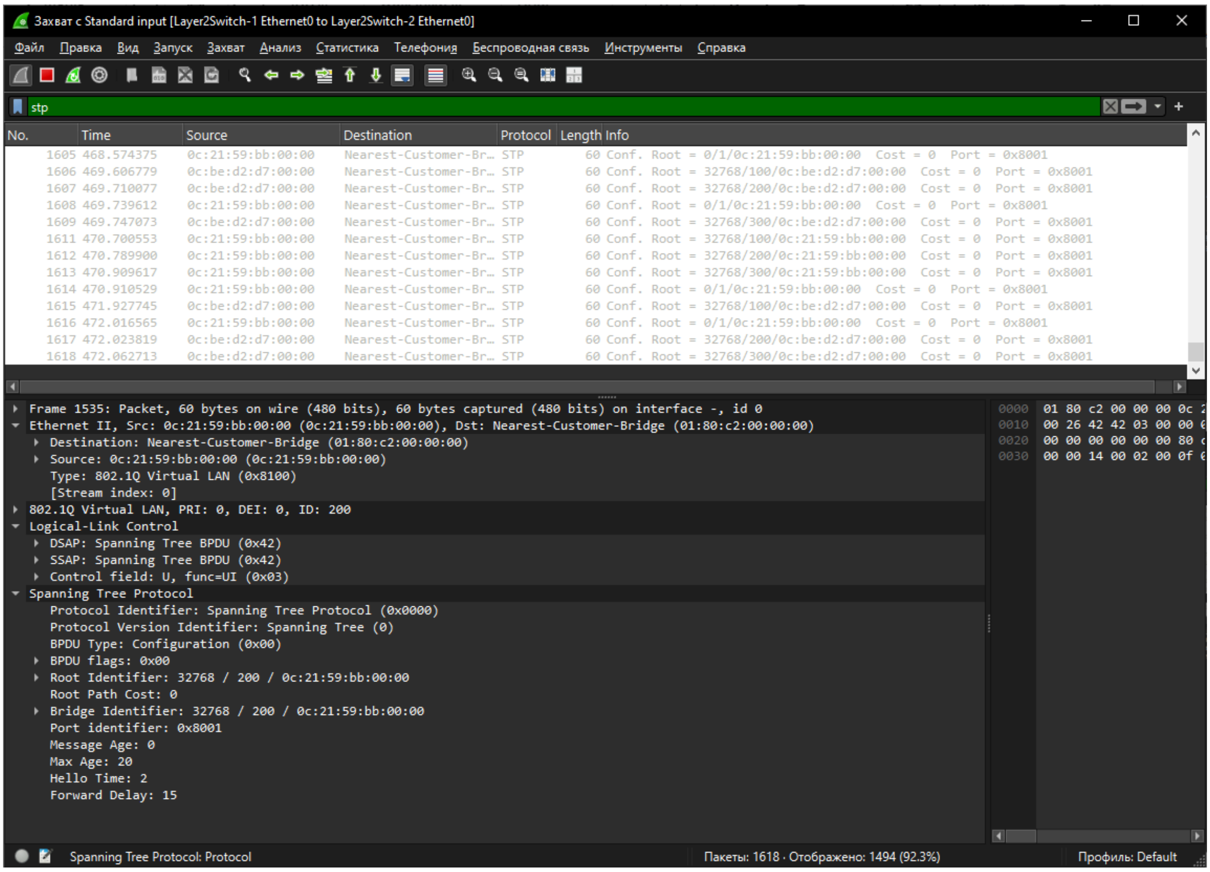
4. Запустил захват на всех линках

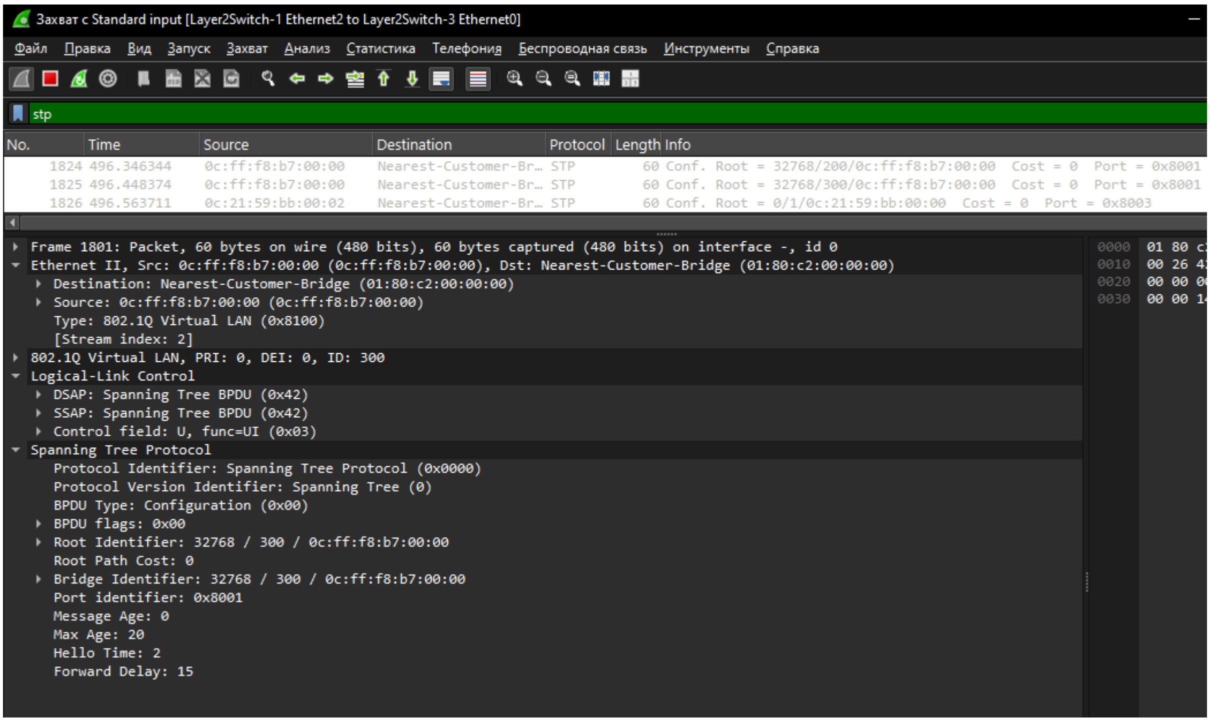


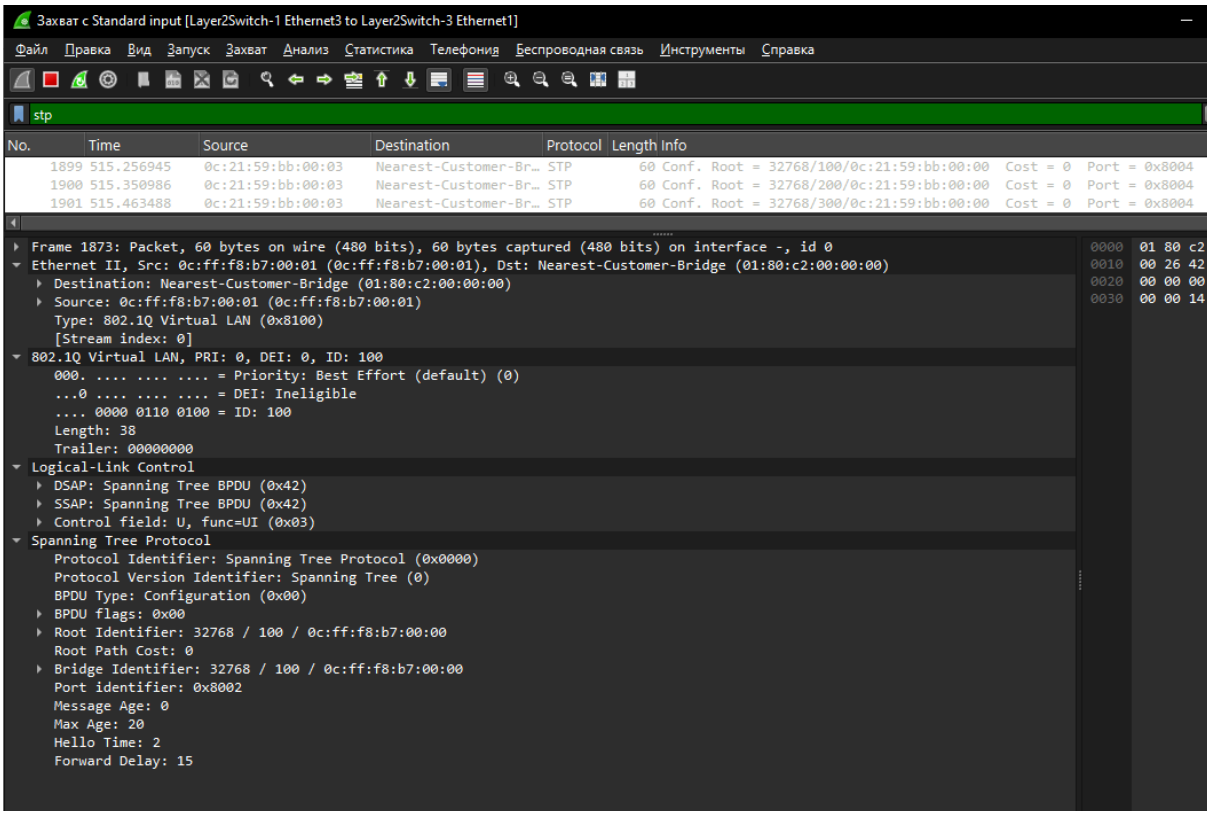
Скриншоты захваченных пакетов с фильтром stp(идут по порядку)

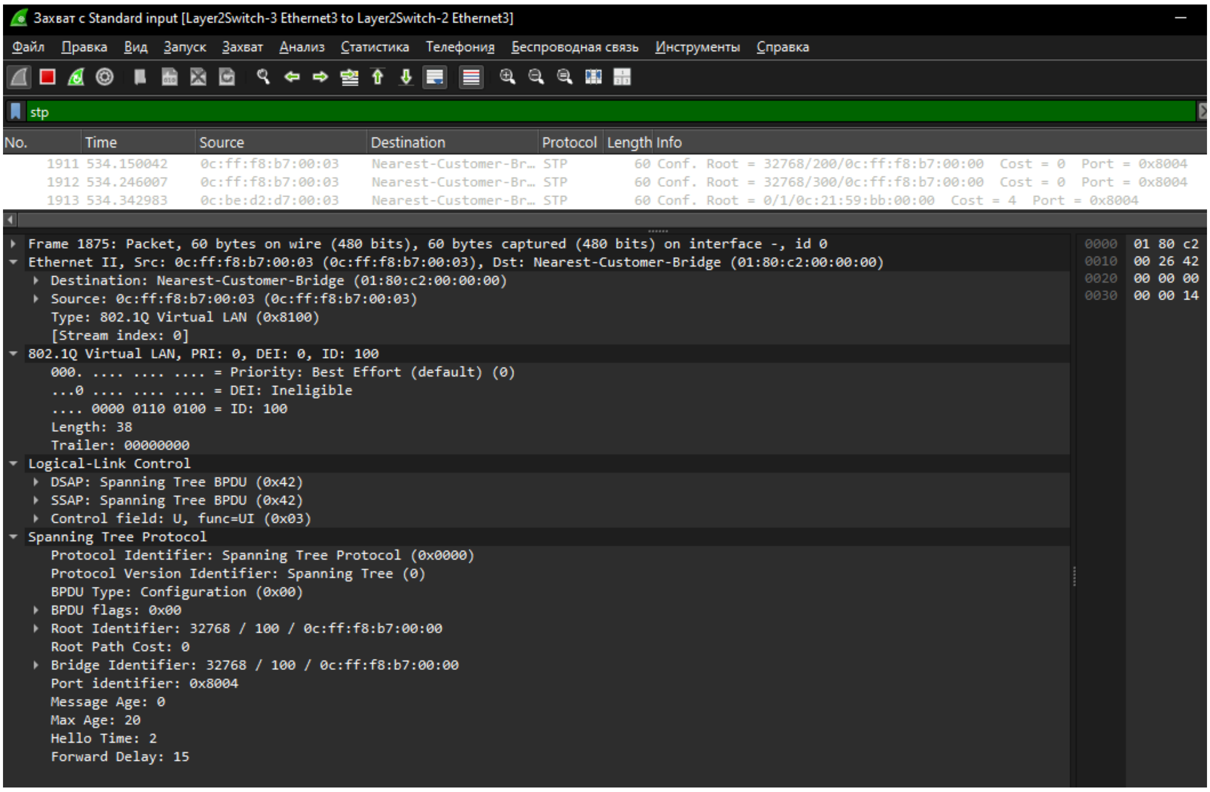

Захват с Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]
Захват с Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]
Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]
Захват с Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]
Захват с Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]
Захват с Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]
Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]
Захват с Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]
Захват с Standard input [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]
Захват с Standard input [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]
Захват с Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]
Захват с Standard input [Layer2Switch-2 Ethernet5 to Layer2Switch-4 Ethernet3]
Захват с Standard input [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]
Захват с Standard input [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]
Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]
Захват с Standard input [Layer2Switch-3 Ethernet5 to PC2 Ethernet0]
Захват с Standard input [Layer2Switch-4 Ethernet4 to PC3 Ethernet0]
Захват с Standard input [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]
Захват с Standard input [Layer2Switch-5 Ethernet4 to PC5 Ethernet0]
Захват с Standard input [Layer2Switch-5 Ethernet5 to PC6 Ethernet0]

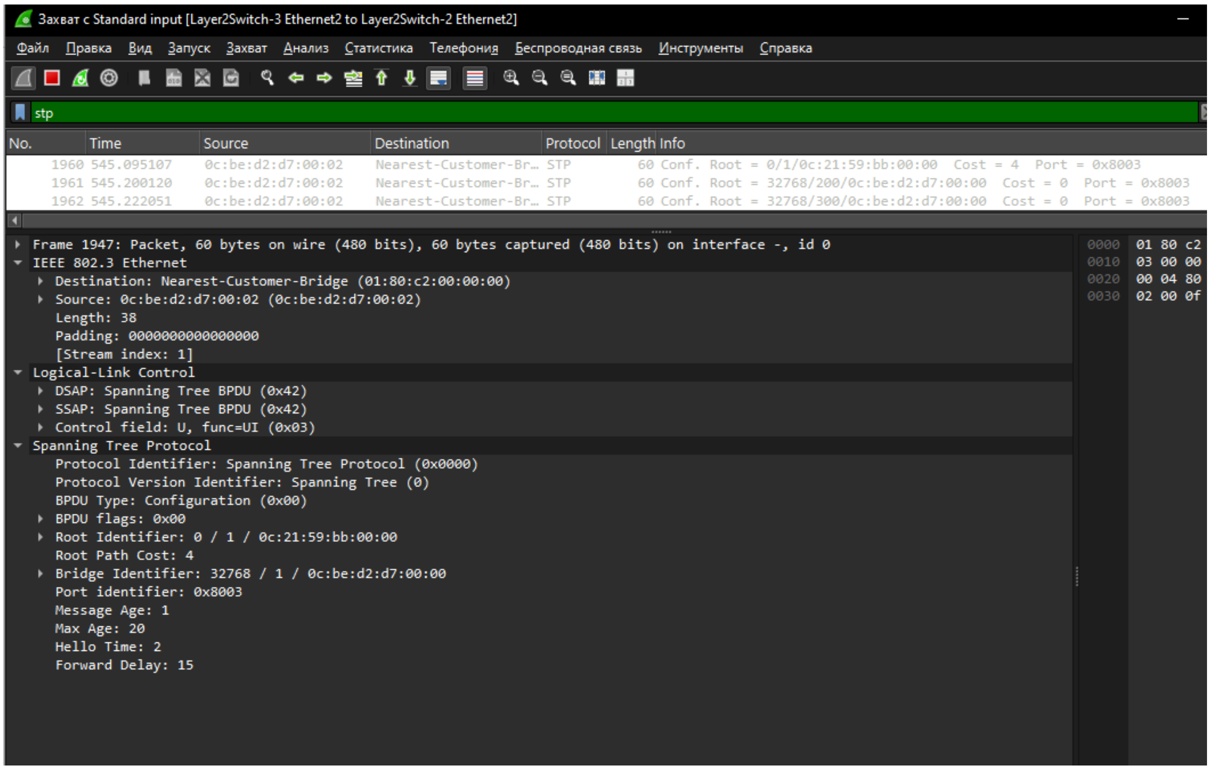

Захват с Standard input [Layer2Switch-1 Ethernet1 to Layer2Switch-2 Ethernet1]
stp
Frame 981: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:21:59:bb:00:01 (0c:21:59:bb:00:01)
Length: 38
Padding: 0000000000000000
[Stream index: 1]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 0
Bridge Identifier: 0 / 1 / 0c:21:59:bb:00:00
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15

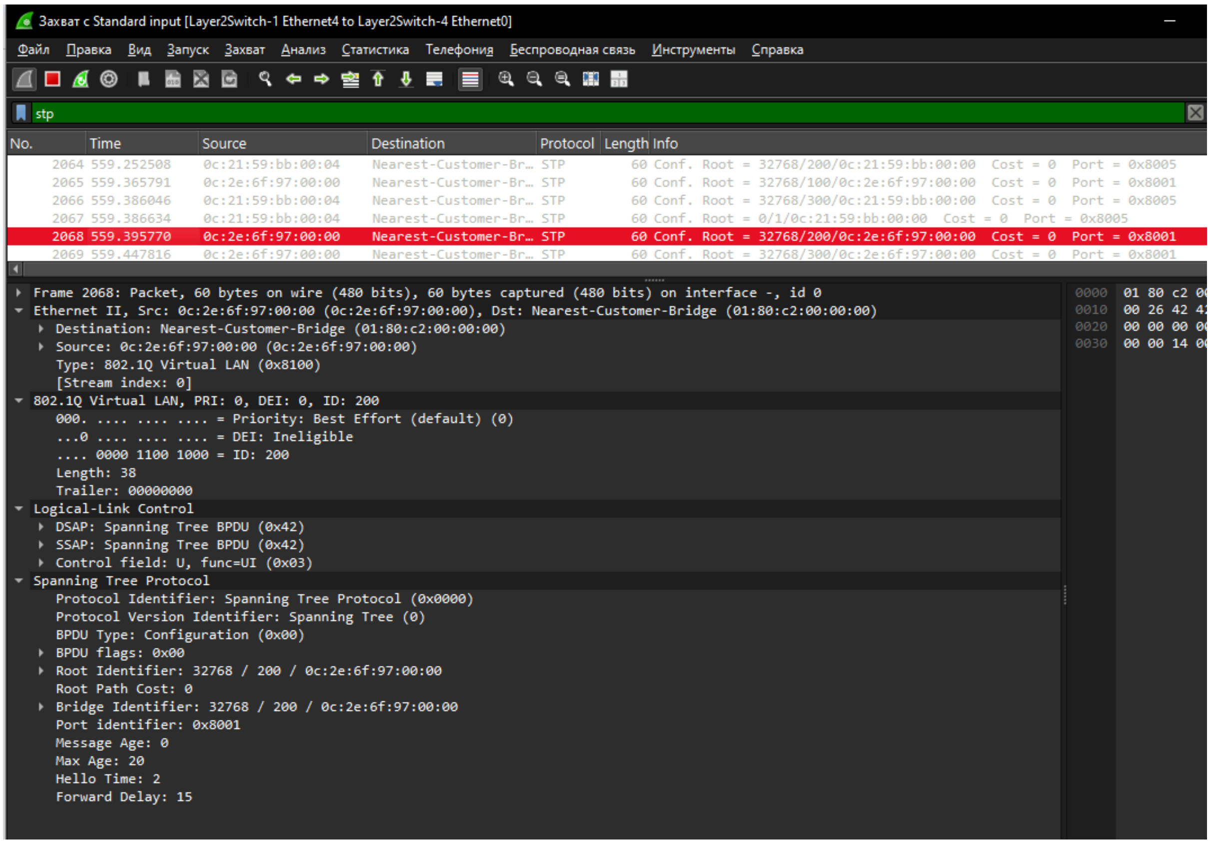

Захват с Standard input [Layer2Switch-1 Ethernet0 to Layer2Switch-2 Ethernet0]
stp
Frame 1535: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:21:59:bb:00:00 (0c:21:59:bb:00:00), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:21:59:bb:00:00 (0c:21:59:bb:00:00)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 200 / 0c:21:59:bb:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 200 / 0c:21:59:bb:00:00
Port identifier: 0x8001
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
Spanning Tree Protocol: Protocol
Пакеты: 1618 · Отображено: 1494 (92.3%)
Профиль: Default

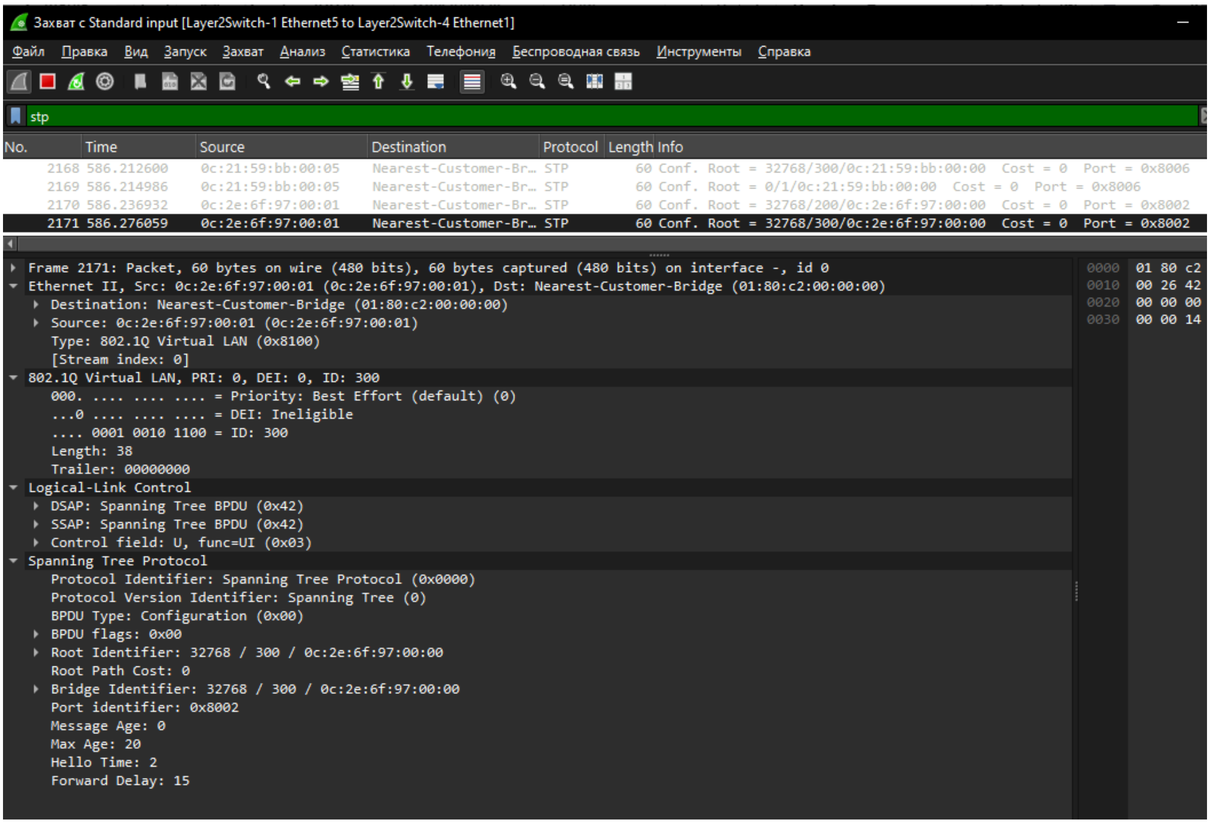

Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
1824 496.346344 0c:ff:f8:b7:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:ff:f8:b7:00:00 Cost = 0 Port = 0x8001
1825 496.448374 0c:ff:f8:b7:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:ff:f8:b7:00:00 Cost = 0 Port = 0x8001
1826 496.563711 0c:21:59:bb:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 0 Port = 0x8003
Frame 1801: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:ff:f8:b7:00:00 (0c:ff:f8:b7:00:00), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:ff:f8:b7:00:00 (0c:ff:f8:b7:00:00)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 2]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 300 / 0c:ff:f8:b7:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 300 / 0c:ff:f8:b7:00:00
Port identifier: 0x8001
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15

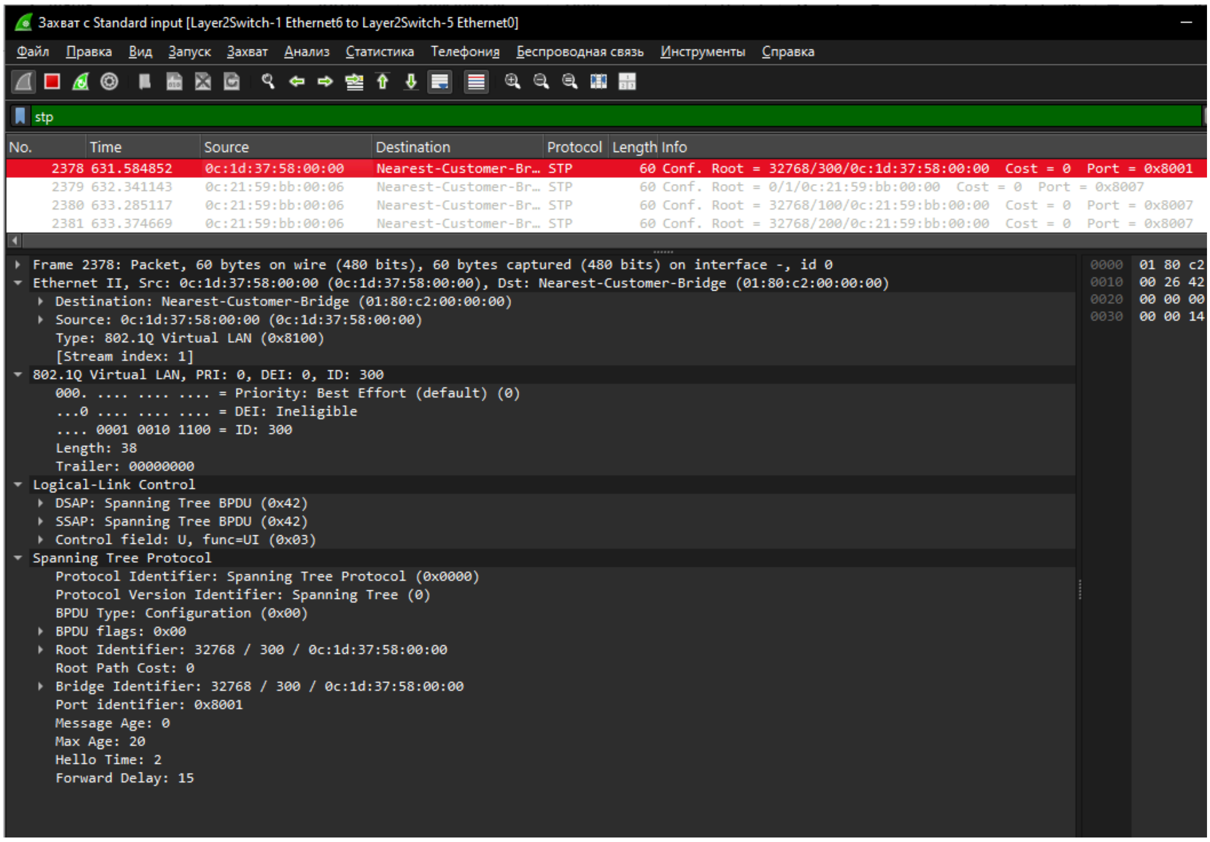

Захват с Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
1899 515.256945 0c:21:59:bb:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:21:59:bb:00:00 Cost = 0 Port = 0x8004
1900 515.350986 0c:21:59:bb:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:21:59:bb:00:00 Cost = 0 Port = 0x8004
1901 515.463488 0c:21:59:bb:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:21:59:bb:00:00 Cost = 0 Port = 0x8004
Frame 1873: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:ff:f8:b7:00:01 (0c:ff:f8:b7:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:ff:f8:b7:00:01 (0c:ff:f8:b7:00:01)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0000 0110 0100 = ID: 100
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 100 / 0c:ff:f8:b7:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 100 / 0c:ff:f8:b7:00:00
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15

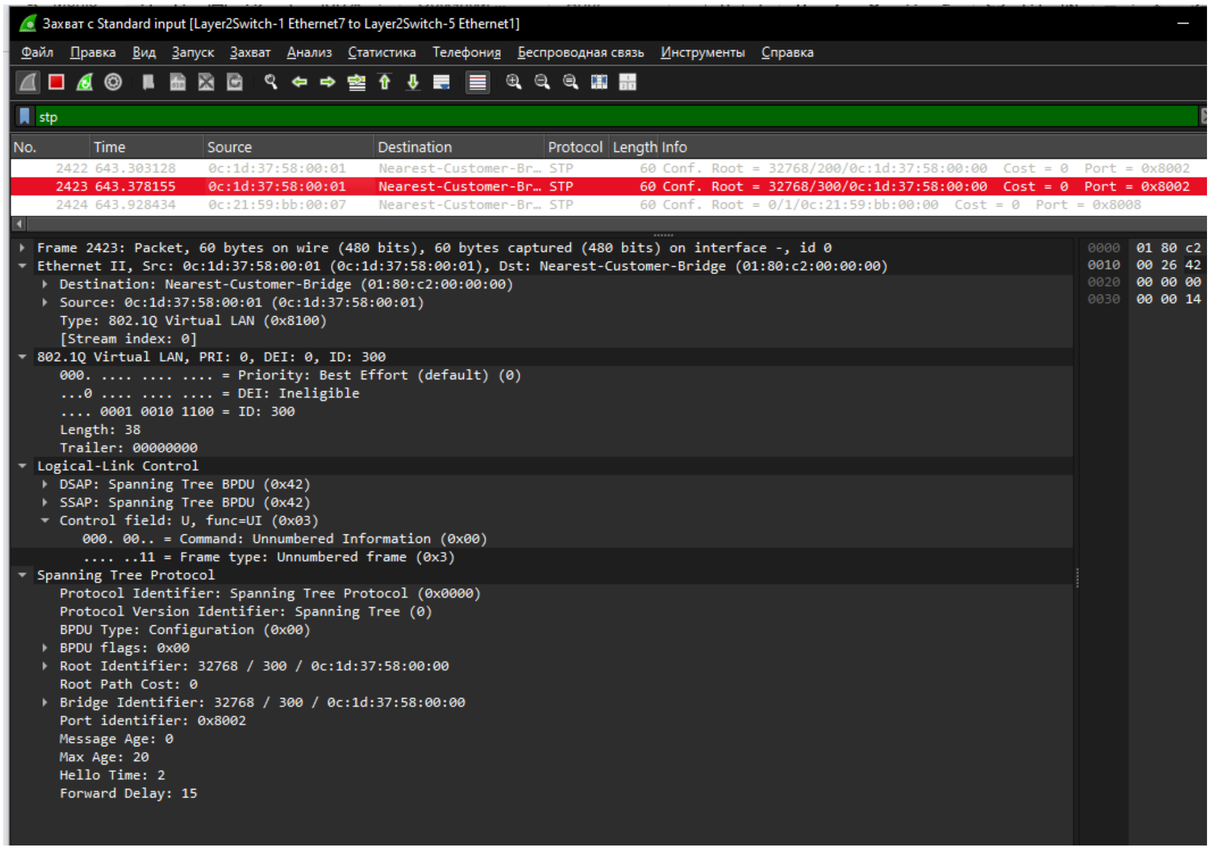

Захват с Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
1911 534.150042 0c:ff:f8:b7:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:ff:f8:b7:00:00 Cost = 0 Port = 0x8004
1912 534.246007 0c:ff:f8:b7:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:ff:f8:b7:00:00 Cost = 0 Port = 0x8004
1913 534.342983 0c:be:d2:d7:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8004
Frame 1875: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:ff:f8:b7:00:03 (0c:ff:f8:b7:00:03), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:ff:f8:b7:00:03 (0c:ff:f8:b7:00:03)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0000 0110 0100 = ID: 100
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 100 / 0c:ff:f8:b7:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 100 / 0c:ff:f8:b7:00:00
Port identifier: 0x8004
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2
0010 00 26 42
0020 00 00 00
0030 00 00 14

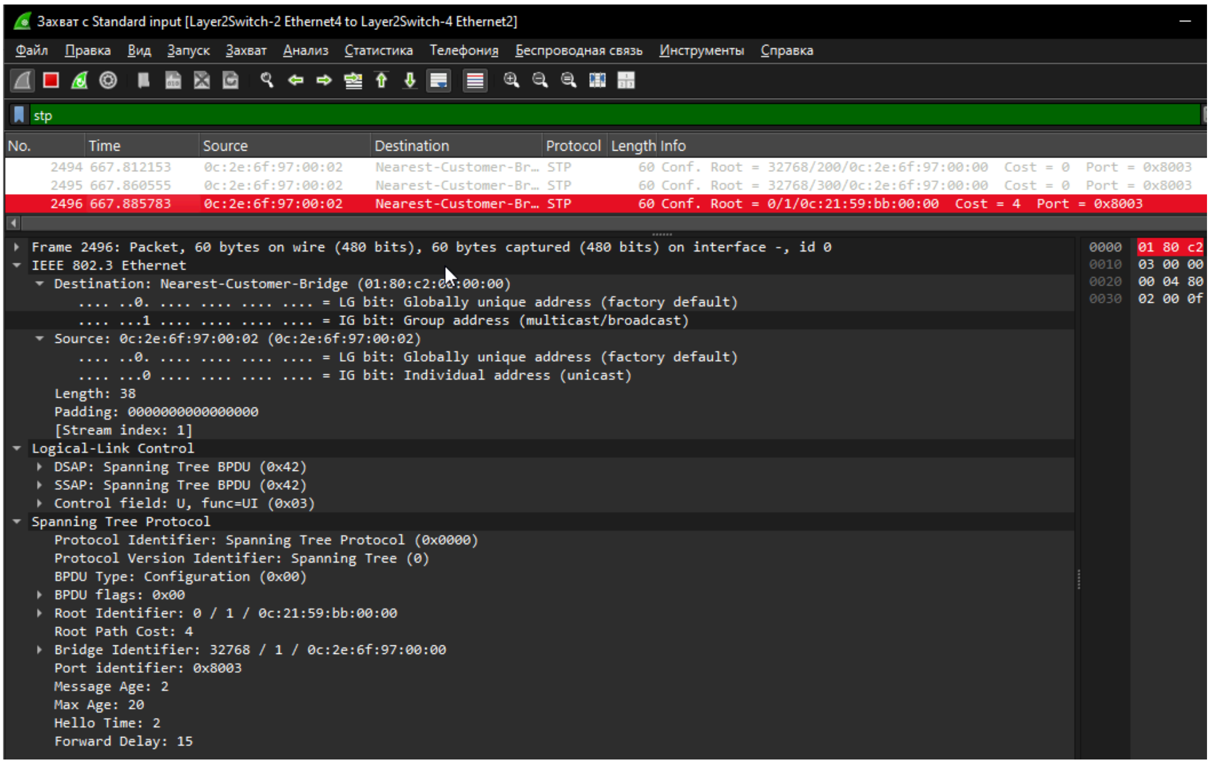

Захват с Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
1960 545.095107 0c:be:d2:d7:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8003
1961 545.200120 0c:be:d2:d7:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8003
1962 545.222051 0c:be:d2:d7:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8003
Frame 1947: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:be:d2:d7:00:02 (0c:be:d2:d7:00:02)
Length: 38
Padding: 0000000000000000
[Stream index: 1]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 4
Bridge Identifier: 32768 / 1 / 0c:be:d2:d7:00:00
Port identifier: 0x8003
Message Age: 1
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2
0010 03 00 00
0020 00 04 80
0030 02 00 0f

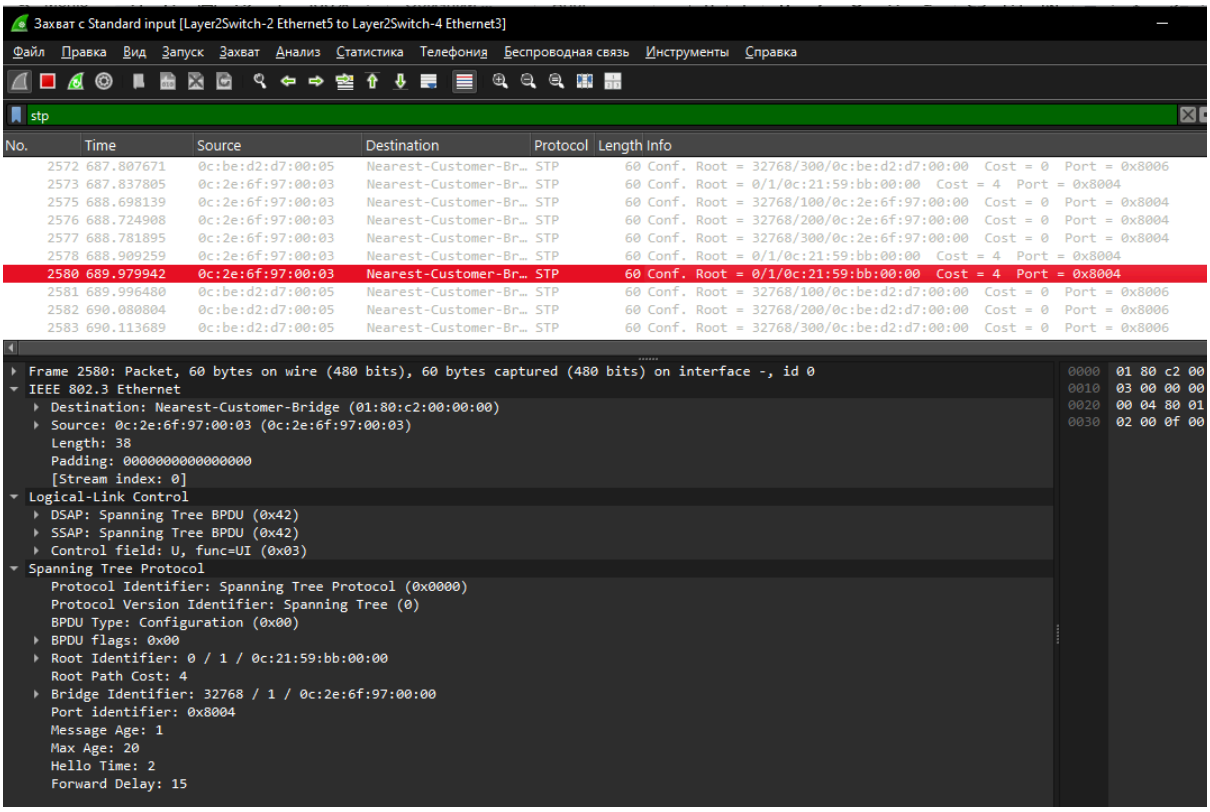

Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2064 559.252508 0c:21:59:bb:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:21:59:bb:00:00 Cost = 0 Port = 0x8005
2065 559.365791 0c:2e:6f:97:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:2e:6f:97:00:00 Cost = 0 Port = 0x8001
2066 559.386046 0c:21:59:bb:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:21:59:bb:00:00 Cost = 0 Port = 0x8005
2067 559.386634 0c:21:59:bb:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 0 Port = 0x8005
2068 559.395770 0c:2e:6f:97:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:2e:6f:97:00:00 Cost = 0 Port = 0x8001
2069 559.447816 0c:2e:6f:97:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:2e:6f:97:00:00 Cost = 0 Port = 0x8001
Frame 2068: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:2e:6f:97:00:00 (0c:2e:6f:97:00:00), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:2e:6f:97:00:00 (0c:2e:6f:97:00:00)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0000 1100 1000 = ID: 200
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 200 / 0c:2e:6f:97:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 200 / 0c:2e:6f:97:00:00
Port identifier: 0x8001
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15

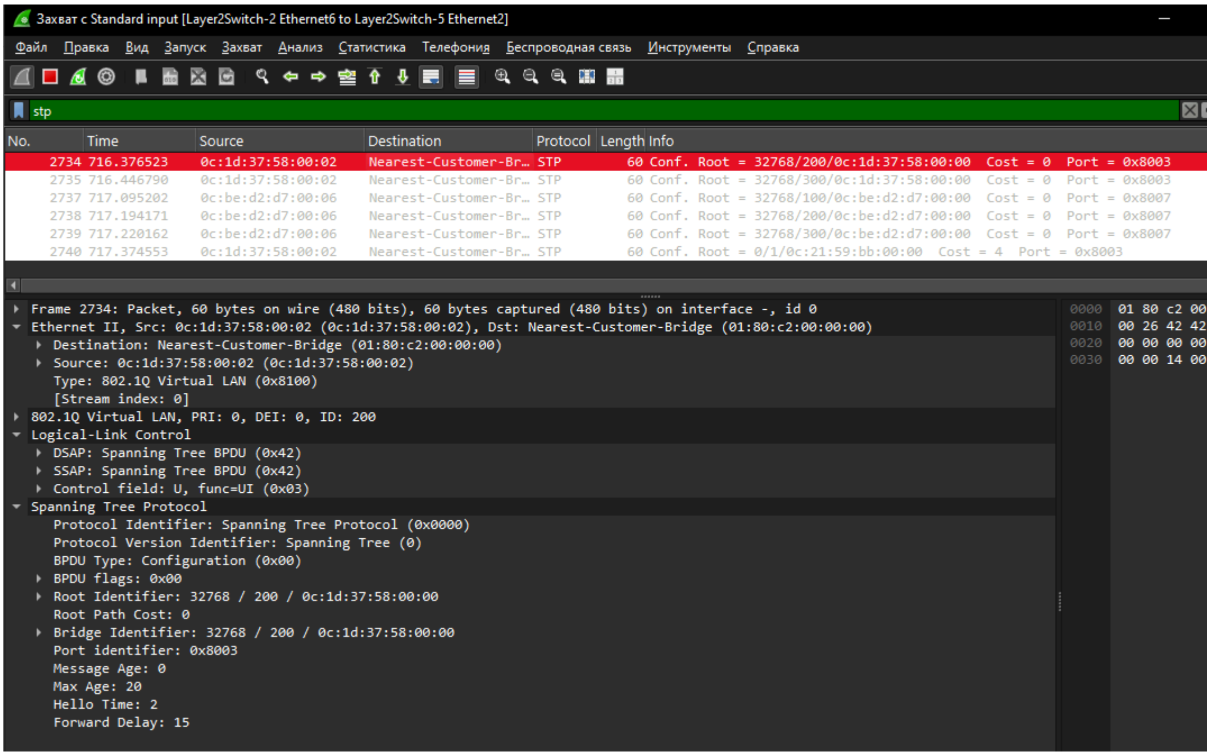

Захват с Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2168 586.212600 0c:21:59:bb:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:21:59:bb:00:00 Cost = 0 Port = 0x8006
2169 586.214986 0c:21:59:bb:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 0 Port = 0x8006
2170 586.236932 0c:2e:6f:97:00:01 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:2e:6f:97:00:00 Cost = 0 Port = 0x8002
2171 586.276059 0c:2e:6f:97:00:01 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:2e:6f:97:00:00 Cost = 0 Port = 0x8002
Frame 2171: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:2e:6f:97:00:01 (0c:2e:6f:97:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:2e:6f:97:00:01 (0c:2e:6f:97:00:01)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0001 0010 1100 = ID: 300
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 300 / 0c:2e:6f:97:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 300 / 0c:2e:6f:97:00:00
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15

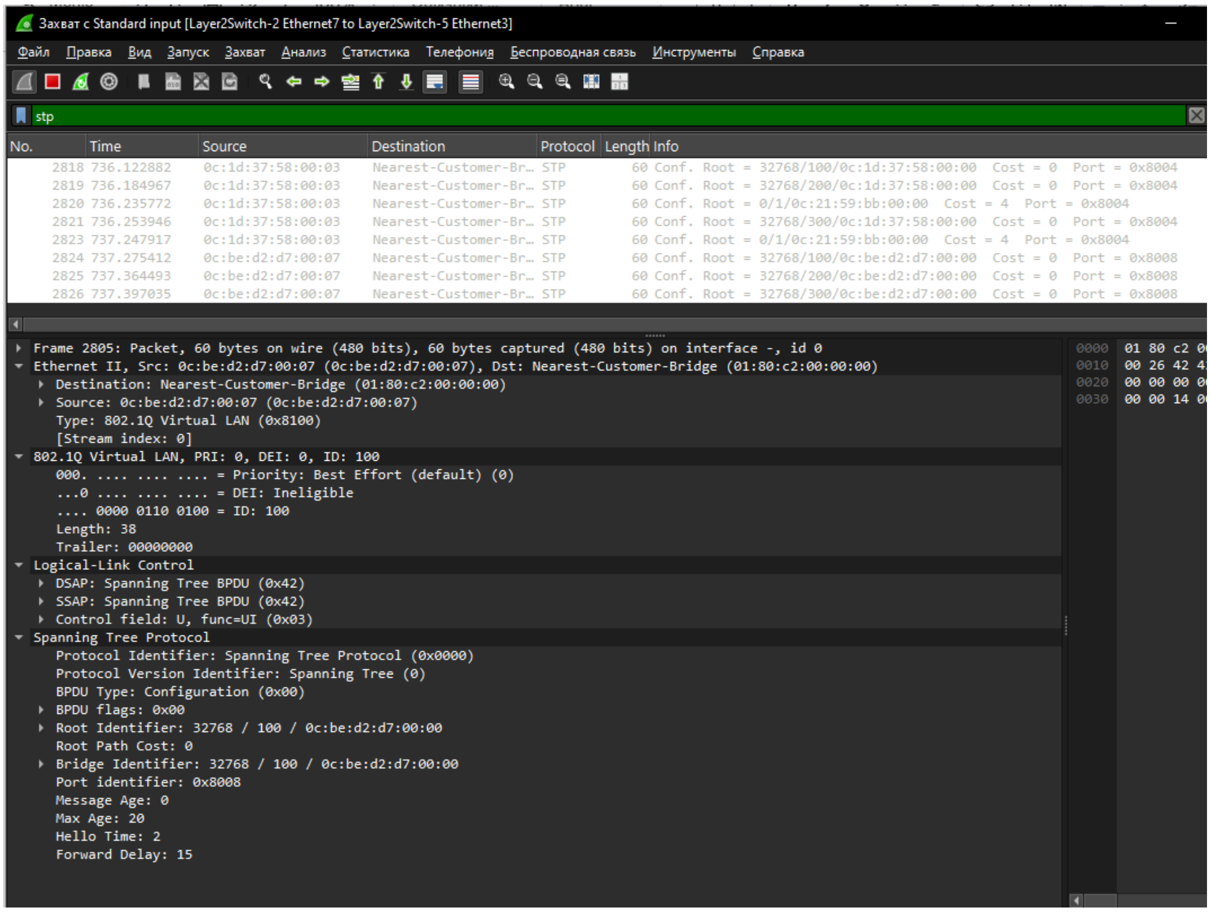

Захват с Standard input [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2378 631.584852 0c:1d:37:58:00:00 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:1d:37:58:00:00 Cost = 0 Port = 0x8001
2379 632.341143 0c:21:59:bb:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 0 Port = 0x8007
2380 633.285117 0c:21:59:bb:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:21:59:bb:00:00 Cost = 0 Port = 0x8007
2381 633.374669 0c:21:59:bb:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:21:59:bb:00:00 Cost = 0 Port = 0x8007
Frame 2378: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:1d:37:58:00:00 (0c:1d:37:58:00:00), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:1d:37:58:00:00 (0c:1d:37:58:00:00)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 1]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0001 0010 1100 = ID: 300
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 300 / 0c:1d:37:58:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 300 / 0c:1d:37:58:00:00
Port identifier: 0x8001
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2
0010 00 26 42
0020 00 00 00
0030 00 00 14

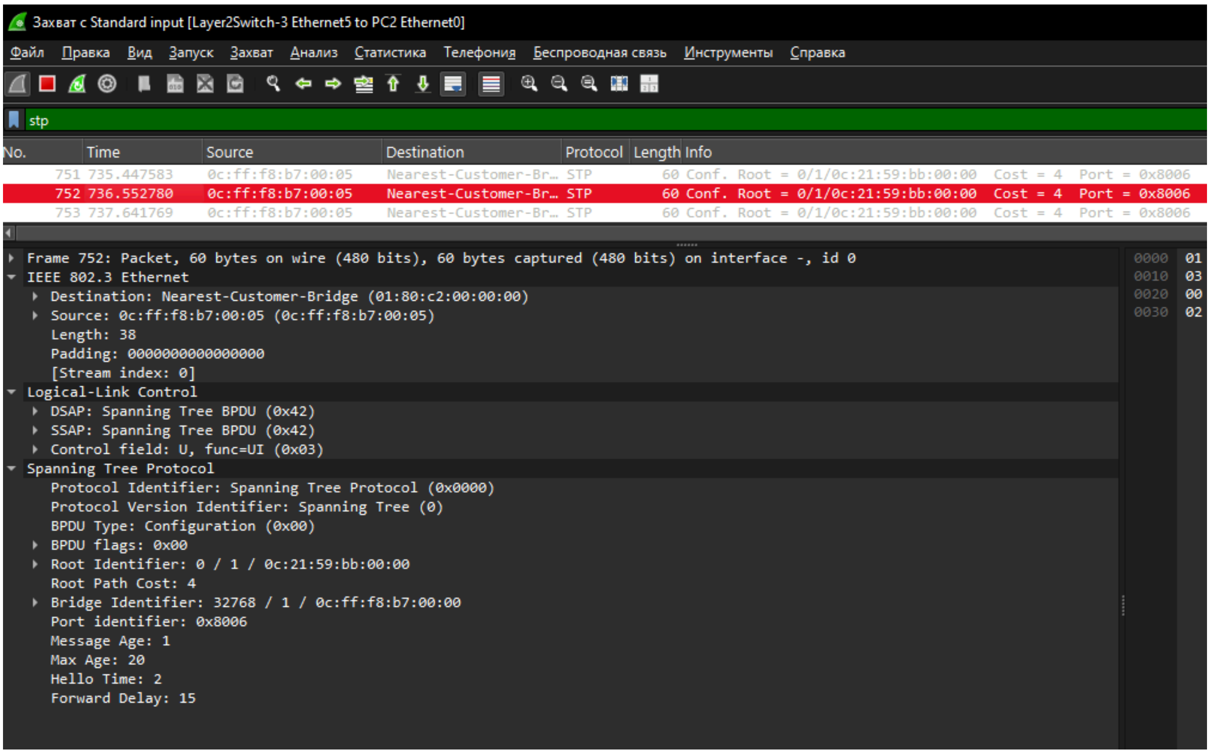

Захват с Standard input [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2422 643.303128 0c:1d:37:58:00:01 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:1d:37:58:00:00 Cost = 0 Port = 0x8002
2423 643.378155 0c:1d:37:58:00:01 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:1d:37:58:00:00 Cost = 0 Port = 0x8002
2424 643.928434 0c:21:59:bb:00:07 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 0 Port = 0x8008
Frame 2423: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:1d:37:58:00:01 (0c:1d:37:58:00:01), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:1d:37:58:00:01 (0c:1d:37:58:00:01)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 300
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0001 0010 1100 = ID: 300
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
000. 00.. = Command: Unnumbered Information (0x00)
.... ..11 = Frame type: Unnumbered frame (0x3)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 300 / 0c:1d:37:58:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 300 / 0c:1d:37:58:00:00
Port identifier: 0x8002
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2
0010 00 26 42
0020 00 00 00
0030 00 00 14

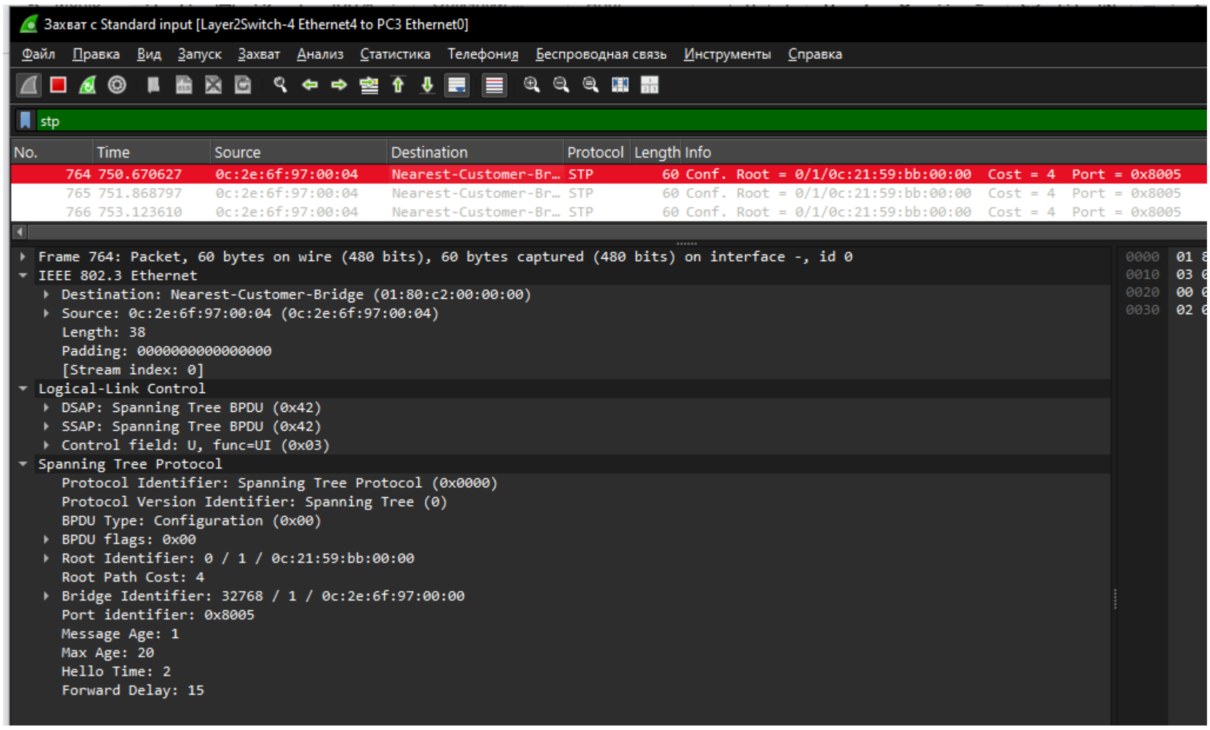


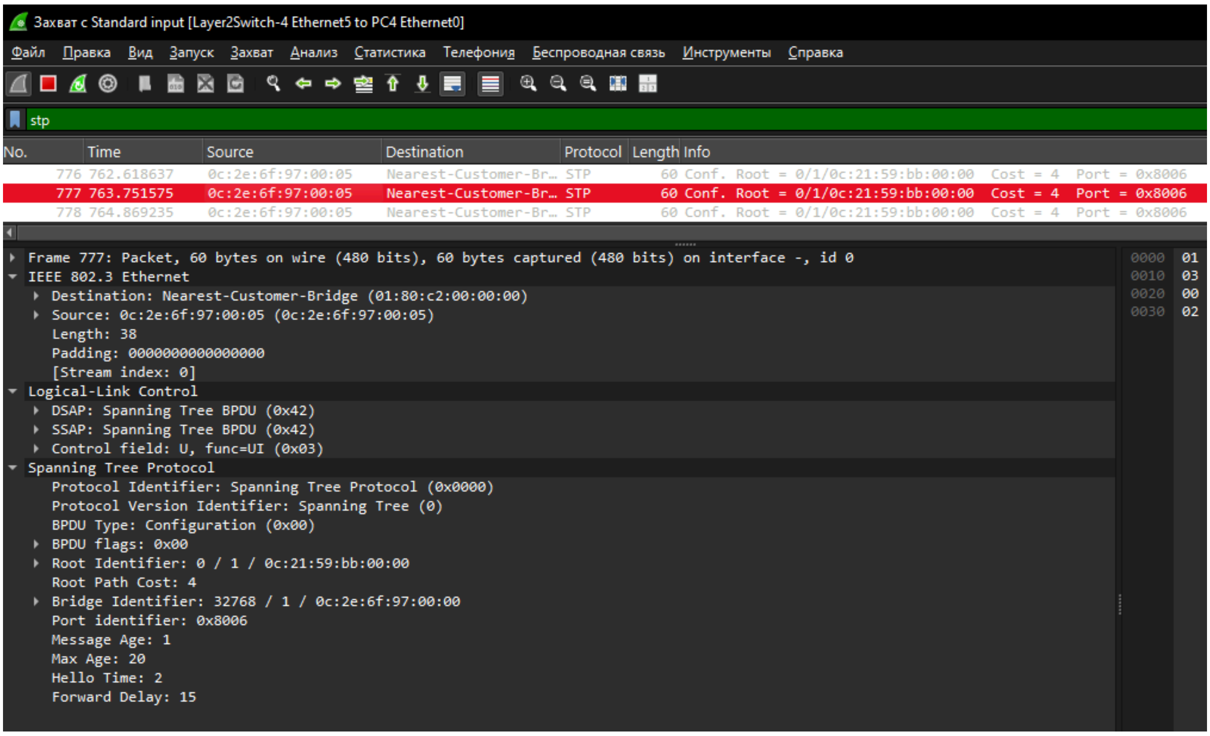


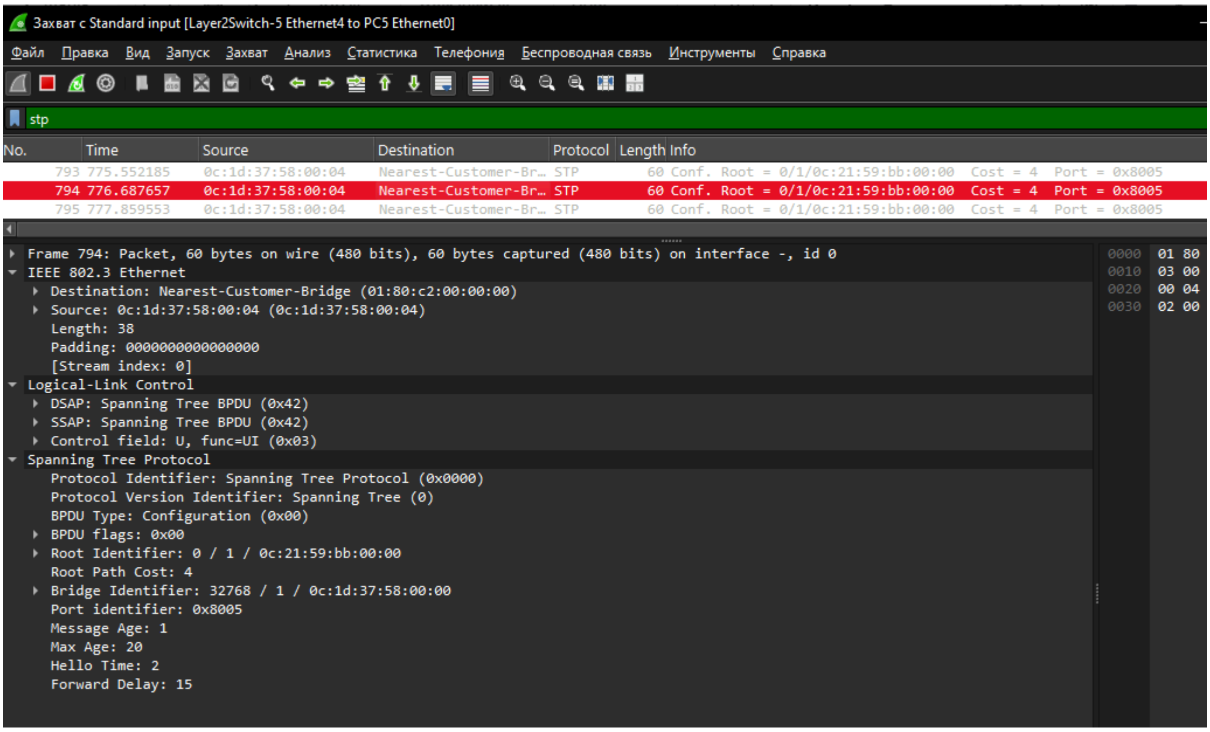

Захват с Standard input [Layer2Switch-2 Ethernet6 to Layer2Switch-5 Ethernet2]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2734 716.376523 0c:1d:37:58:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:1d:37:58:00:00 Cost = 0 Port = 0x8003
2735 716.446790 0c:1d:37:58:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:1d:37:58:00:00 Cost = 0 Port = 0x8003
2737 717.095202 0c:be:d2:d7:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8007
2738 717.194171 0c:be:d2:d7:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8007
2739 717.220162 0c:be:d2:d7:00:06 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8007
2740 717.374553 0c:1d:37:58:00:02 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8003
Frame 2734: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:1d:37:58:00:02 (0c:1d:37:58:00:02), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:1d:37:58:00:02 (0c:1d:37:58:00:02)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 200
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 200 / 0c:1d:37:58:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 200 / 0c:1d:37:58:00:00
Port identifier: 0x8003
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2 00
0010 00 26 42 42
0020 00 00 00 00
0030 00 00 14 00

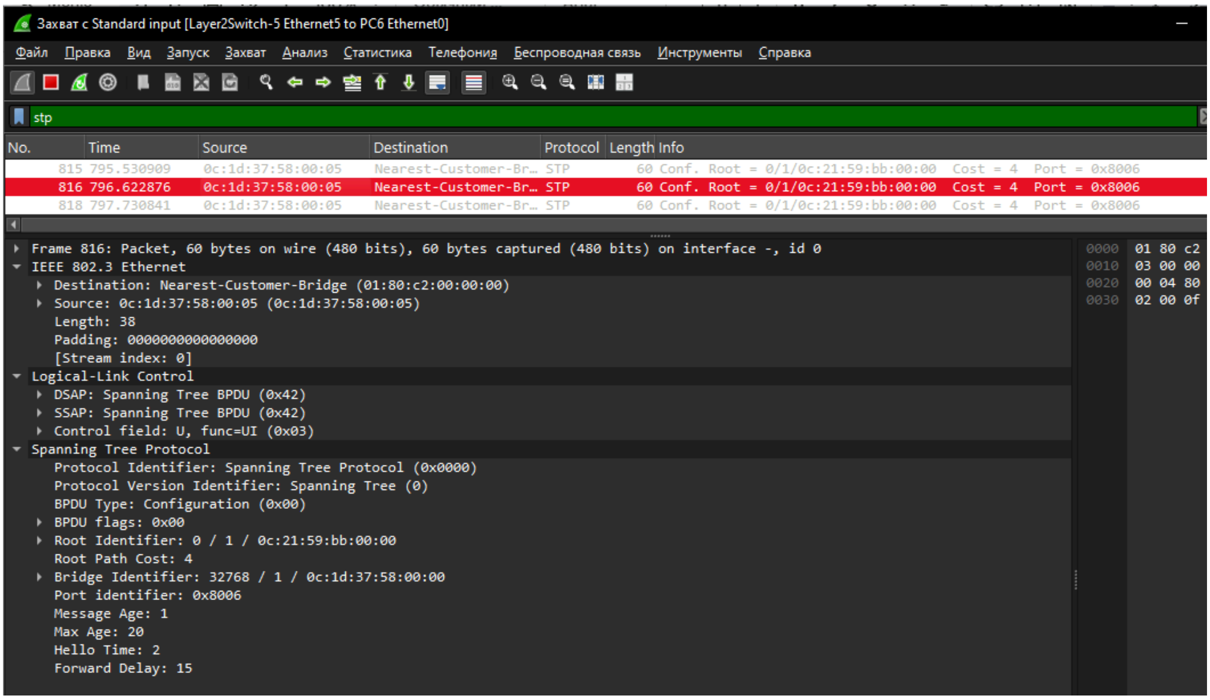

Захват с Standard input [Layer2Switch-2 Ethernet7 to Layer2Switch-5 Ethernet3]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
2818 736.122882 0c:1d:37:58:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:1d:37:58:00:00 Cost = 0 Port = 0x8004
2819 736.184967 0c:1d:37:58:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:1d:37:58:00:00 Cost = 0 Port = 0x8004
2820 736.235772 0c:1d:37:58:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8004
2821 736.253946 0c:1d:37:58:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:1d:37:58:00:00 Cost = 0 Port = 0x8004
2823 737.247917 0c:1d:37:58:00:03 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8004
2824 737.275412 0c:be:d2:d7:00:07 Nearest-Customer-Br… STP 60 Conf. Root = 32768/100/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8008
2825 737.364493 0c:be:d2:d7:00:07 Nearest-Customer-Br… STP 60 Conf. Root = 32768/200/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8008
2826 737.397035 0c:be:d2:d7:00:07 Nearest-Customer-Br… STP 60 Conf. Root = 32768/300/0c:be:d2:d7:00:00 Cost = 0 Port = 0x8008
Frame 2805: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
Ethernet II, Src: 0c:be:d2:d7:00:07 (0c:be:d2:d7:00:07), Dst: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:be:d2:d7:00:07 (0c:be:d2:d7:00:07)
Type: 802.1Q Virtual LAN (0x8100)
[Stream index: 0]
802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 100
000. .... = Priority: Best Effort (default) (0)
...0 .... = DEI: Ineligible
.... 0000 0110 0100 = ID: 100
Length: 38
Trailer: 00000000
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 32768 / 100 / 0c:be:d2:d7:00:00
Root Path Cost: 0
Bridge Identifier: 32768 / 100 / 0c:be:d2:d7:00:00
Port identifier: 0x8008
Message Age: 0
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80 c2
0010 00 26 42
0020 00 00 00
0030 00 00 14


Захват с Standard input [Layer2Switch-3 Ethernet5 to PC2 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
751 735.447583 0c:ff:f8:b7:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
752 736.552780 0c:ff:f8:b7:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
753 737.641769 0c:ff:f8:b7:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
Frame 752: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:ff:f8:b7:00:05 (0c:ff:f8:b7:00:05)
Length: 38
Padding: 0000000000000000
[Stream index: 0]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 4
Bridge Identifier: 32768 / 1 / 0c:ff:f8:b7:00:00
Port identifier: 0x8006
Message Age: 1
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01
0010 03
0020 00
0030 02


Захват с Standard input [Layer2Switch-4 Ethernet4 to PC3 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
764 750.670627 0c:2e:6f:97:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
765 751.868797 0c:2e:6f:97:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
766 753.123610 0c:2e:6f:97:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
Frame 764: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:2e:6f:97:00:04 (0c:2e:6f:97:00:04)
Length: 38
Padding: 0000000000000000
[Stream index: 0]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 4
Bridge Identifier: 32768 / 1 / 0c:2e:6f:97:00:00
Port identifier: 0x8005
Message Age: 1
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000
0010
0020
0030

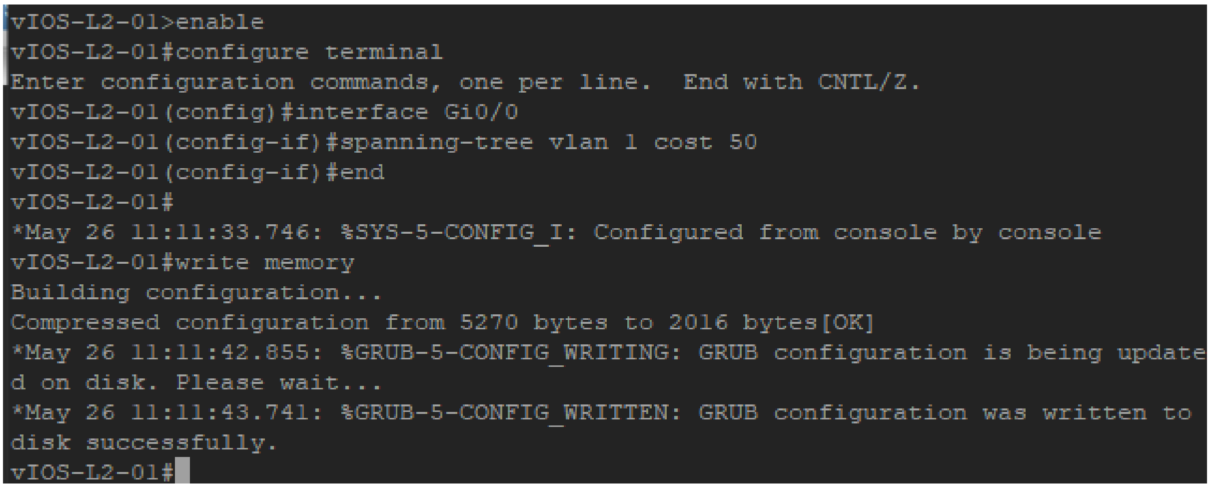

Захват с Standard input [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
776 762.618637 0c:2e:6f:97:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
777 763.751575 0c:2e:6f:97:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
778 764.869235 0c:2e:6f:97:00:05 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8006
Frame 777: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:2e:6f:97:00:05 (0c:2e:6f:97:00:05)
Length: 38
Padding: 0000000000000000
[Stream index: 0]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 4
Bridge Identifier: 32768 / 1 / 0c:2e:6f:97:00:00
Port identifier: 0x8006
Message Age: 1
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01
0010 03
0020 00
0030 02


Захват с Standard input [Layer2Switch-5 Ethernet4 to PC5 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
stp
No. Time Source Destination Protocol Length Info
793 775.552185 0c:1d:37:58:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
794 776.687657 0c:1d:37:58:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
795 777.859553 0c:1d:37:58:00:04 Nearest-Customer-Br… STP 60 Conf. Root = 0/1/0c:21:59:bb:00:00 Cost = 4 Port = 0x8005
Frame 794: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0
IEEE 802.3 Ethernet
Destination: Nearest-Customer-Bridge (01:80:c2:00:00:00)
Source: 0c:1d:37:58:00:04 (0c:1d:37:58:00:04)
Length: 38
Padding: 0000000000000000
[Stream index: 0]
Logical-Link Control
DSAP: Spanning Tree BPDU (0x42)
SSAP: Spanning Tree BPDU (0x42)
Control field: U, func=UI (0x03)
Spanning Tree Protocol
Protocol Identifier: Spanning Tree Protocol (0x0000)
Protocol Version Identifier: Spanning Tree (0)
BPDU Type: Configuration (0x00)
BPDU flags: 0x00
Root Identifier: 0 / 1 / 0c:21:59:bb:00:00
Root Path Cost: 4
Bridge Identifier: 32768 / 1 / 0c:1d:37:58:00:00
Port identifier: 0x8005
Message Age: 1
Max Age: 20
Hello Time: 2
Forward Delay: 15
0000 01 80
0010 03 00
0020 00 04
0030 02 00



5. Изменил стоимость пути на SW2

enable
configure terminal
interface Gi0/0
spanning-tree vlan 1 cost 50
end
write memory



```
vIOS-L2-01>enable
vIOS-L2-01#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
vIOS-L2-01(config)#interface Gi0/0
vIOS-L2-01(config-if)#spanning-tree vlan 1 cost 50
vIOS-L2-01(config-if)#end
vIOS-L2-01#
*May 26 11:11:33.746: %SYS-5-CONFIG_I: Configured from console by console
vIOS-L2-01#write memory
Building configuration...
Compressed configuration from 5270 bytes to 2016 bytes[OK]
*May 26 11:11:42.855: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 11:11:43.741: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

пишу show spanning-tree vlan 1 на всех свичах

sw1

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    1      (priority 0 sys-id-ext 1)
             Address     0c21.59bb.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------------
Gi0/0               Desg FWD 4         128.1    Shr
Gi0/1               Desg FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
Gi1/2               Desg FWD 4         128.7    Shr
Gi1/3               Desg FWD 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

SW2(изменный cost)

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        2 (GigabitEthernet0/1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cbe.d2d7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------------------------
Gi0/0               Altn BLK 50        128.1    Shr
Gi0/1               Root FWD 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Altn BLK 4         128.5    Shr
Gi1/1               Altn BLK 4         128.6    Shr
Gi1/2               Altn BLK 4         128.7    Shr
Gi1/3               Altn BLK 4         128.8    Shr
Gi2/0               Desg FWD 4         128.9    Shr
```

SW3

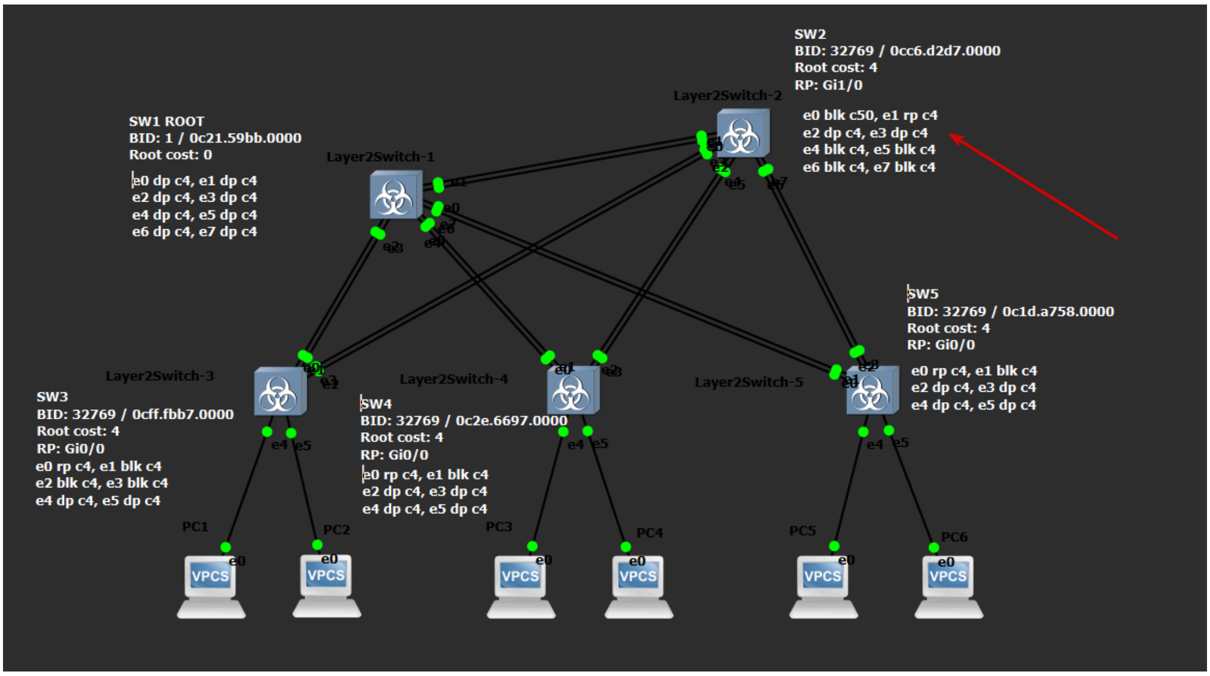
```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0cff.f8b7.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Altn BLK 4         128.3    Shr
Gi0/3               Altn BLK 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

SW4

```
vIOS-L2-01>
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c2e.6f97.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

```
vIOS-L2-01>enable
vIOS-L2-01#show spanning-tree vlan 1

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    1
             Address     0c21.59bb.0000
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0c1d.3758.0000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ------------------------
Gi0/0               Root FWD 4         128.1    Shr
Gi0/1               Altn BLK 4         128.2    Shr
Gi0/2               Desg FWD 4         128.3    Shr
Gi0/3               Desg FWD 4         128.4    Shr
Gi1/0               Desg FWD 4         128.5    Shr
Gi1/1               Desg FWD 4         128.6    Shr
```

На SW2 бывший root порт стал altn, и STP провел перерасчет, теперь root Gi0/1



6. На каждом свитче

enable
copy running-config startup-config
terminal length 0
show running-config

SW1

```
vIOS-L2-01#QQ
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5274 bytes to 2022 bytes[OK]
*May 26 11:52:43.138: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 11:52:43.978: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

```
vIOS-L2-01#show running-config
Building configuration...

Current configuration : 5274 bytes
!
! Last configuration change at 10:28:40 UTC Tue May 26 2026
!
version 15.0
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname vIOS-L2-01
!
boot-start-marker
boot-end-marker
!
!
!
no aaa new-model
!
!
!
!
!
vtp domain CISCO-vIOS
vtp mode transparent
!
!
!
ip cef
no ipv6 cef
!
```

SW2

```
vIOS-L2-01>enable
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5270 bytes to 2016 bytes[OK]
vIOS-L2-01#
*May 26 11:35:53.067: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 11:35:53.956: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

SW3

```
vIOS-L2-01>enable
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5042 bytes to 1934 bytes[OK]
vIOS-L2-01#
*May 26 11:57:24.049: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 11:57:24.888: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

SW4

```
vIOS-L2-01>enable
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5042 bytes to 1934 bytes[OK]
vIOS-L2-01#
*May 26 11:57:46.418: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 11:57:47.259: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

SW5

```
vIOS-L2-01>enable
vIOS-L2-01#
vIOS-L2-01#
vIOS-L2-01#
vIOS-L2-01#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
Compressed configuration from 5042 bytes to 1934 bytes[OK]
*May 26 12:03:05.792: %GRUB-5-CONFIG_WRITING: GRUB configuration is being update
d on disk. Please wait...
*May 26 12:03:06.618: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to
disk successfully.
vIOS-L2-01#
```

На каждом PC
show ip
save
PC1

```
PC1> show ip

NAME        : PC1[1]
IP/MASK     : 192.168.1.1/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:00
LPORT       : 21812
RHOST:PORT  : 127.0.0.1:21813
MTU         : 1500

PC1> save
Saving startup configuration to startup.vpc
.  done
```

PC2

```
PC2> show ip

NAME        : PC2[1]
IP/MASK     : 192.168.1.2/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:01
LPORT       : 21814
RHOST:PORT  : 127.0.0.1:21815
MTU         : 1500

PC2> save
Saving startup configuration to startup.vpc
.  done

PC2>
```

PC3

```
PC3> show ip

NAME        : PC3[1]
IP/MASK     : 192.168.1.3/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:02
LPORT       : 21816
RHOST:PORT  : 127.0.0.1:21817
MTU         : 1500

PC3> save
Saving startup configuration to startup.vpc
.  done

PC3>
```

PC4

```
PC4> show ip

NAME         : PC4[1]
IP/MASK      : 192.168.1.4/24
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:03
LPORT        : 21818
RHOST:PORT   : 127.0.0.1:21819
MTU          : 1500

PC4> save
Saving startup configuration to startup.vpc
.  done

PC4>
```

PC5

```
PC5> show ip

NAME         : PC5[1]
IP/MASK      : 192.168.1.5/24
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:04
LPORT        : 21820
RHOST:PORT   : 127.0.0.1:21821
MTU          : 1500

PC5> save
Saving startup configuration to startup.vpc
.  done

PC5>
```

PC6

```
PC6> show ip

NAME         : PC6[1]
IP/MASK      : 192.168.1.6/24
GATEWAY      : 0.0.0.0
DNS          :
MAC          : 00:50:79:66:68:05
LPORT        : 21822
RHOST:PORT   : 127.0.0.1:21823
MTU          : 1500

PC6> save
Saving startup configuration to startup.vpc
.  done

PC6>
```

Текстовые файлы с конфигурациями сохранены в configs/*.txt